

PROmpter — Manual

PROmpter automatically turns the export files of the **LightCycler® PRO** into a formatted **PDF** and **Excel overview** (for later LIS processing) — an independent, private project.

Important: PROmpter is a free, **non-validated helper tool for display**. It takes over Cq values, instrument calls (`finalCall`) and curves **unchanged from the LightCycler PRO** and summarises the calls per sample clearly by a **fixed rule** (see "How PROmpter summarises (transparently)"); it does not recalculate any values and sets no thresholds of its own. The reports are a **results overview** — the results in the instrument software and the **result release by the laboratory** remain authoritative. PROmpter is **not placed on the market as a medical device or in-vitro diagnostic**, carries **no CE marking** and has **not been formally assessed from a regulatory standpoint**. Before routine use, it must be **checked and released within the laboratory's quality management**. Liability is governed by the statutory provisions. PROmpter is **initially intended for qualitative analyses** — i.e. the **reactive / non-reactive** call per target, as the LightCycler PRO provides it in `finalCall`. **Quantitative** results (absolute or relative quantification via standard curves) are **currently not displayed**.

In a nutshell

1. In the **Settings**, choose a **watched folder** (where the LC PRO drops its export files).
2. **Start monitoring** — done. Every new file is processed automatically.
3. For each file a **PDF** and an **Excel** are created; the **original** is then moved to the archive (or deleted — depending on the setting).

PROmpter may run in the background: with **"Work in background"** it sits as an icon in the notification area (tray) next to the clock, and monitoring stays active. The **window's close button ()**, in contrast, **quits PROmpter completely**.

Installation & first start

PROmpter does **not** need to be installed — it is a single `.exe`. It is best to copy it to a **fixed local location** (e.g. `C:\PROmpter\`) and start it by double-clicking; a desktop shortcut makes it convenient. **Administrator rights are not required** — neither on the first nor on later starts.

On the very first start Windows may show a blue message **"Windows protected your PC"** (Microsoft Defender SmartScreen). This is **not** an error and **not** a virus, just a notice because the program is not (yet) digitally signed. Here is how to continue:

1. Click **"More info"**.
2. Then click **"Run anyway"**.

This is only necessary **once per program version** — after that PROmpter starts without asking. (If the file is in a OneDrive folder, the message may reappear; at a local location such as `C:\PROmpter\` it stays a one-time thing.)

Note for clinical/laboratory IT

PROmpter is a **stand-alone, portable** Windows application (Python/PySide6/Qt) **without an installer, without administrator rights and without a network service**. It reads only the configured **watched folder** and writes **PDF/Excel** to the chosen output folders, plus its settings to `%APPDATA%\PROmpter\`. The **only** optional outgoing network access is a **version check** when clicking `""` (fetching a small text file; can be disabled by leaving out the update URL).

So that staff do not see the SmartScreen message at all, IT can **whitelist** the file — e.g. via a SmartScreen exception or through AppLocker/WDAC. The cleanest long-term solution is to **sign** the `.exe` with a **code-signing certificate** of the institution; then the warning disappears entirely.

Exporting the run at the LightCycler PRO

For PROmpter to process a run, it must be exported at the LightCycler PRO as **customer data** (a ZIP file with XML). Here is how at the device:

1. Open the **Results** application and tick the **completed run** in the **run overview**.
2. Choose the **Export** button.
3. In the "**Select location for file and export**" dialog, choose the option "**Export customer data**".

Important: do *not* choose "Perform data export" — that creates an **LCRD** file (folder run_data). That file is **encrypted** and **cannot** be read by PROmpter at all. PROmpter needs the open **customer data** (ZIP/XML).

4. Choose the **location** (SFTP server or USB device) and confirm with **Export**.

The customer data is stored as a **ZIP archive** (named after the device serial number) in the customer_data folder; inside it there is one XML per run (named after run ID + timestamp). You set **exactly this target folder** (or the folder into which the SFTP transfer lands) as the **watched folder** in PROmpter — then every new export is automatically processed into PDF + Excel.

Tip: Several completed runs can also be **exported together** — PROmpter then creates a separate report for **each run** (see "Several runs in one file").

The main window

- **Start/Stop monitoring** — switches the automatic folder monitoring on/off.
- **Process file now ...** — select and process a single export file by hand.
- **Scan folder now** — search the watched folder once for files not yet processed.
- **Settings ...** — all options (see below).
- **Work in background** — places the window into the notification area (tray); monitoring keeps running in the background (bring it back via the tray icon's "Show", "Quit" closes it completely).
- **Close** — the **window's close button ()** quits PROmpter **completely** (not to the notification area).
- **Log** — shows for each file the time, name, status (**green** = OK, **yellow** = notice/partially processed, **red** = error) and details. If something fails, **the original file stays put** — nothing is lost.
- **Save log ...** — save the displayed lines as **CSV** (for Excel) or as a **text file**.
- **Clear log** — clears only the display; the PDF/Excel files already created are kept.

Settings

The settings are organised in four tabs: **File management** (folders, which files, original action, printing, file names), **PDF report** (sections, header/footer, sorting, curve type), **Excel report** (columns, sorting, sheets, curve type) and **CSV/TXT export** (Roche-style format, columns, delimiter). **Sorting and curve type can be set separately per report** — the PDF may e.g. be by column and the Excel by row.

For **sorting the samples** there are four options: **by column** (A1, B1, C1 ...), **by row** (A1, A2, A3 ...) as well as **alphabetically ascending or descending** by the **sample name**.

Sprache / Language

At the very top of the **File management** tab, the **language** can be changed: **Deutsch** (default) or **English**. The change affects the **interface**, the **PDF report**, the **Excel file** and **all messages**; in English mode numbers are written with a **point** instead of a comma (e.g. Cq 15 . 62). **Device and sample data** (assay/sample names, Cq,

finalCall, curves) stay unchanged in **both** languages. The new language takes effect **after a restart** of the program. *(The manual is available in German and English and opens automatically in the language that is set.)*

Folders

- **Watched folder** — is monitored automatically.
- **PDF output / Excel output** — target folders. **Empty = same folder as the export file** (the finished PDF/Excel are not read in again).
- **Archive folder** — where the original is moved to. Empty = subfolder Archiv/. (Deliberately a **separate** folder so that the moved original is not immediately processed again as a "new file".)

Original file after processing

- **Move to archive** (default) or **delete**. *"Leave in place" deliberately does not exist*, so that the same run is not accidentally processed twice.

Printing

PROMptter can **automatically print the PDF report after it has been created** — handy for the paper archive of an unattended watched folder.

- The **checkbox "Print PDF report automatically after creation"** turns the function on (default: off).
- **Printer** — the **default printer** (Windows) or a specific printer from the list of installed printers. This lets the monitoring PC print specifically to the **lab-report printer**.
- **Copies** — number of printouts per report (e.g. 2 for a duplicate archive).
- **PROMptter prints the PDF itself** directly to the chosen printer — **regardless of which PDF program** is installed or set as default on the PC (no Adobe/Edge needed).
- **Safety: only the generated PDF** is ever printed, never the original file. If printing fails once (e.g. no printer reachable), the run is still **counted as processed** (PDF/Excel are saved after all) and marked **yellow** with a notice in the log.

Sorting the samples

- **by column** (A1, B1, C1 ...) or **by row** (A1, A2, A3 ...) — selectable **separately** in the *PDF report* and *Excel report* tabs.
- **Controls (NTC/PTC)** always appear in the Excel, clearly marked as a control. In the **summary** (Excel) and the **Cq table** (PDF) they are additionally highlighted: **PTC red, NTC green, reactive samples bold**.

Curve display

Which curve form computed by the device is displayed (selectable **separately** in the *PDF report* and *Excel report* tabs). In brackets the name that the **LightCycler PRO** itself uses for it:

Selection (PROMptter)	At the device	Meaning
Baseline divided (default)	"Baseline divided" (bdcurve)	Each curve point is divided by the baseline . This is what the LC PRO always computes its results from — hence the most meaningful display.
Baseline subtracted	"Baseline subtracted" (bscurve)	The baseline is subtracted from each point .
Raw values	"Raw data" / "Fluorescence history"	The unprocessed measured fluorescence.

The **baseline** is the y-axis intercept of an amplification curve; each curve has its own. All three forms come **unchanged from the device** — PROMptter does not compute its own baseline. *(Terms as per the LightCycler PRO user assistance.)*

Two curve views (each individually switchable on/off in PDF and Excel):

- **per sample** — a small chart per well with all channels (overview per patient sample); the targets/channels are coloured as a **spectrum blue -> dark red**.
- **per target** — *one* chart per channel/target, in which **all samples lie on top of each other** (the classic amplification-curve view). The curves are **coloured by result** (reactive coloured, negative grey, controls highlighted); reactive curves and controls are **numbered**, with a small **mapping table** next to them (no. -> well -> sample -> Cq). In Excel, each sample is additionally its own line whose name appears **on mouse-over**. The **Y scale is aligned per target to the strongest sample** (the fluorescence level differs from target to target).

File names of the reports

The name of the generated PDF/Excel is assembled from **building blocks**. Buttons insert placeholders; the **preview** immediately shows the result. (*The placeholders are inserted by buttons and are the same in both languages.*)

Placeholder	Content
{Plattenname}	Name of the plate setup, e.g. "260101_Example_Assay"
{Datum}	Run date, e.g. 2026-03-27
{DatumZeit}	Date + time
{Uhrzeit}	Time
{LaufID}	Run ID
{PlateID}	Plate ID
{LCAP}	1st LCAP of the run, e.g. "Assay_A" (the old name {Assay} still works)
{Profil}	PCR profile, e.g. "Profile_1"
{LCAPVersion}	Version of the 1st LCAP, e.g. "1.0" (the old name {AssayVersion} still works)
{Geraet}	Device serial number
{Bearbeiter}	Operator from the file

Free text is allowed (e.g. Result_{Datum}_{LaufID}). Empty blocks and invalid characters are cleaned up automatically; if the name is the same, PROmpter appends _2, _3 ... and **never overwrites**.

Header and footer of the PDF

- **Header** — up to **two freely chosen lines** (e.g. "Example University" / "Virus diagnostics"). Below it the **plate name - run ID** is shown.
- **Footer (left)** — freely composed from **building blocks** (the same placeholders as for the file name), default {Datum}_{PlateID}_{LaufID}. The **page number** stays on the right.

PDF — which sections appear

Header, control-validity check, plate overview, result overview, individual values (Cq), **amplification curves per sample** and **amplification curves per target** — each section individually switchable on/off.

Excel — which sheets appear

As with the PDF sections, a checkbox can be set per **sheet**: **Result table**, **Summary**, **Amplification curves per sample**, **Amplification curves per target**, **Reactive samples**.

Excel — which columns appear

A checkbox per column (for the "Results" sheet). The table is built "long" (one row per sample × target) — good for the later LIS import.

The outputs

PDF report

Header with run key data · control-validity check (NTC/PTC) · coloured **plate overview (96 or 384 wells, automatically detected)** · **result overview per sample** (overall statement, reactive targets with Cq, internal control) · **individual values** (Cq per target) · **amplification curves per sample** (one chart per well, all channels) · **amplification curves per target** (all samples per channel overlaid, coloured by result + mapping table). Uniform Y scale **from 0**, aligned to the samples.

Excel workbook

- **Results** — the configured columns (flags in plain text). The **"Flags"** column shows **all** flags, the separately selectable **"Critical flags"** column only the heavily weighted ones.
- **Summary** — cross table: one row per sample, one column per detection target with the Cq (**empty = not tested**, **"-" = tested but negative**), plus internal control, overall result and the **"Critical flags"** column (plain-text reason), plus a tally.
- **Curves** — native, editable Excel line charts per sample.
- **Curves per target** — *one* chart per channel with all samples (coloured by result; sample name on mouse-over).
- **Reactive samples** — the same per-sample charts, only for reactive samples.

CSV/TXT result table (optional, Roche-style)

In addition to PDF and Excel, PROmpter can write a **CSV/TXT result table** per run — handy for colleagues or LIS tools that expect the familiar Roche tables. Enable it in the *File management* tab (checkbox **"CSV/TXT file"**); the format and options are in a dedicated **"CSV/TXT export"** tab. There are **four formats**:

- **LightCycler 96** (default) — one row per well × target (columns *Position · Sample Name · Gene Name · Cq · Call · Sample Type · Dye · EPF ...*), tab-separated, point decimal.
- **LightCycler 480 – classic** — one block per detection target with a title line (*Include · Color · Pos · Name · Cp · ...*), comma decimal.
- **LightCycler 480 – extended** — text columns in quotes (*Experiment Name · Analysis Name · Position · SampleName · CrossingPoint · ... · Call*).
- **Do-it-yourself** — freely choose columns, delimiter (tab/comma/semicolon), decimal separator, quoting, header row and an optional title line.

Important: Cq and the result (Call) come unchanged from the device. PROmpter is qualitative — so the **concentration and standard-curve columns** of the LC480 formats stay **empty or "-"** (the column layout still stays Roche-compatible).

How PROmpter summarises (transparently)

Fixed rule (order of precedence): **reactive > invalid > non-reactive**.

- **reactive** — as soon as **at least one** target is reported by the device as reactive/positive (the affected targets are named) — even if another target is "Invalid".
- **invalid** — no target reactive, but **at least one** target reported by the device as "Invalid".
- **non-reactive** — all detection targets are reported by the device as non-reactive/negative.
- **see individual values** — empty or unexpected call (should practically never occur).
- **Controls (NTC/PTC)** — "control valid" if the device reports all detection targets of the control as "Valid", otherwise "check control". The control check in the PDF only reflects this instrument assessment.
- **Internal control (IC)** — is only **shown**; it does **not** automatically block the result.

- **Flags** are shown in **plain text** (e.g. "EPF value too low").

This overall statement does **not recalculate any values** and sets **no thresholds of its own**; it bundles the device's own individual calls by the fixed rule described above. The results in the instrument software and the result release by the laboratory remain authoritative.

Several runs in one file

If the LightCycler PRO exports **several runs into one file** (one ZIP archive, named after the device serial number), PROMptter creates a separate PDF and a separate Excel for **each run** (named after the respective run). This works **reliably** — so you can safely export several completed runs **together**.

Background: each run is a **self-contained, complete XML file** in the ZIP (named after run ID + timestamp); PROMptter reads them one after another. Should a single run exceptionally be **unreadable** once (e.g. a corrupted export file), PROMptter still processes the remaining runs and marks the file **yellow** in the log with a notice — the whole file is never discarded, and the original is, in this case, **archived instead of deleted** to be safe.

Notes & limitations

- PROMptter is initially designed for **qualitative analyses** (reactive / non-reactive per target). **Quantitative** results (quantification via standard curves) are **currently not** displayed.
 - PROMptter is a **helper tool** and **not validated**; it is not placed on the market as a medical device or in-vitro diagnostic and carries no CE marking (see the notice above). Before routine use, it must be checked and released within the laboratory's quality management. Please check the results, controls and plausibility in the instrument software before every result release.
 - The export files carry `sentToHost=false` / status "unreviewed" — so they are **not yet released**.
 - PROMptter **never** writes to the original data; it only reads and creates new files.
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License & open-source libraries

- You **may** pass PROMptter on to colleagues and use it in the lab (before routine use, after checking and release within the lab's quality management). Private/free distribution is unproblematic.
- PROMptter uses open-source libraries:
- **PySide6/Qt** (© The Qt Company) — license **GNU LGPL v3** · <<https://www.qt.io/licensing>>
- **ReportLab** (PDF generation) — **BSD license** · <<https://www.reportlab.com>>
- **openpyxl** (Excel generation) — **MIT license** · <<https://openpyxl.readthedocs.io>>
- **watchdog** (folder monitoring) — **Apache 2.0 license** · <<https://github.com/gorakhargosh/watchdog>>

These licenses permit the use in PROMptter without disclosing your own source code. As long as the app is passed on **as an unchanged, stand-alone** .exe, all license obligations are met. (You will find the same note in the **/About** dialog of the app.)

- "Roche" and "LightCycler" are trademarks of their respective owners. PROMptter is an **independent, private project**, compatible with LightCycler® PRO exports, and **not** reviewed by or the responsibility of Roche.
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Feedback

Suggestions for improvement and bug reports are welcome — via the **button** in PROMptter (e-mail or online form).